



Signal Detection Theory: Testing Psychology & Parapsychology

Submitted by:

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1 Assignment 1 (50 Points)

Questions

- Compute your friend's d' in a working memory task. You may read out the words, use PowerPoint slides, or program them (e.g., Matlab).
- Does d' vary with set size? Plot d' versus set size — this is called a *psychophysical function*.
- Does bias vary with set size? Interpret your findings from (a) and (b). (30 points)
- Construct an ROC curve and show how it relates to d' . *Hint*: How do you ensure that your subject varies her/his criteria? (20 points)

Useful formulas:

$$d' = \Phi^{-1}(H) - \Phi^{-1}(F) \quad (1)$$

$$c = -\Phi^{-1}(F) \quad (2)$$

$$b = -\frac{\Phi^{-1}(H) + \Phi^{-1}(F)}{2} \quad (3)$$

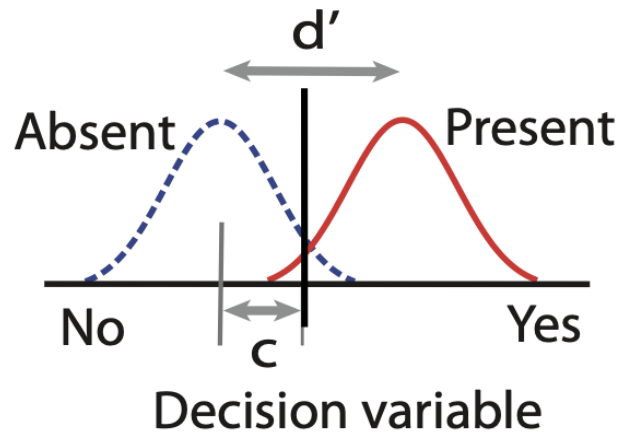


Figure 1: Example of an assignment figure.

Answers

(a) Computation of d' :

An HTML-based application was developed to conduct the experiment. The program automatically saved the results as a .csv file for further analysis and computation. A pool of approximately 150 words was used (listed in Section A), from which a random subset was selected for each trial. The subject was instructed to memorize the displayed words for 5 seconds in each trial. Afterward, a test word was presented, and the subject indicated whether it was part of the original set and rated her confidence in the response on a 5-point scale (1 = Guessing, 2 = Unsure, 3 = Moderate, 4 = Confident, 5 = Certain).

The complete code and recorded results are provided in the Appendix. Informed consent was obtained from the subject (Dr. Gnaani, MBBS, MMST). The experiment was conducted with set sizes ranging from 2 to 15 words, with 15 trials per set, resulting in a total of 165 observations. The obtained dataset is shown below.

Table 1: Experimental results showing Signal Detection Theory parameters (d' , criterion c , bias b , and AUC) derived from participant responses.

Set Size	n_sig	n_noise	Hit Rate	FA Rate	zH	zF	d'	c	b	AUC
2	9	6	0.94	0.08	1.59	-1.38	2.98	-0.11	-0.11	0.98
3	5	10	0.90	0.05	1.28	-1.64	2.93	0.18	0.18	0.98
4	5	10	0.90	0.05	1.28	-1.64	2.93	0.18	0.18	0.98
5	5	10	0.80	0.05	0.84	-1.64	2.49	0.40	0.40	0.96
6	8	7	0.94	0.07	1.53	-1.47	3.00	-0.03	-0.03	0.98
7	9	7	0.94	0.07	1.59	-1.47	3.06	-0.06	-0.06	0.98
8	8	6	0.94	0.08	1.53	-1.38	2.92	-0.08	-0.08	0.98
9	3	12	0.83	0.17	0.97	-0.97	1.93	0.00	0.00	0.91
10	8	7	0.88	0.14	1.15	-1.07	2.22	-0.04	-0.04	0.94
12	5	10	0.90	0.30	1.28	-0.52	1.81	-0.38	-0.38	0.90
15	7	8	0.86	0.63	1.07	0.32	0.75	-0.69	-0.69	0.70

(b) Variation of d' with set size:

Yes, d' (sensitivity) decreased monotonically with set size. The subject exhibited higher discriminability when the memory load was low. As the set size increased, both internal noise and cognitive load rose, leading to a progressive decline in perceptual sensitivity (d'). The psychophysical function shows that d' remained relatively high (~ 3.0) for smaller set sizes (2-7) but declined sharply from set size 9 (≈ 2.0), reaching about 0.75 at the largest set size (15).

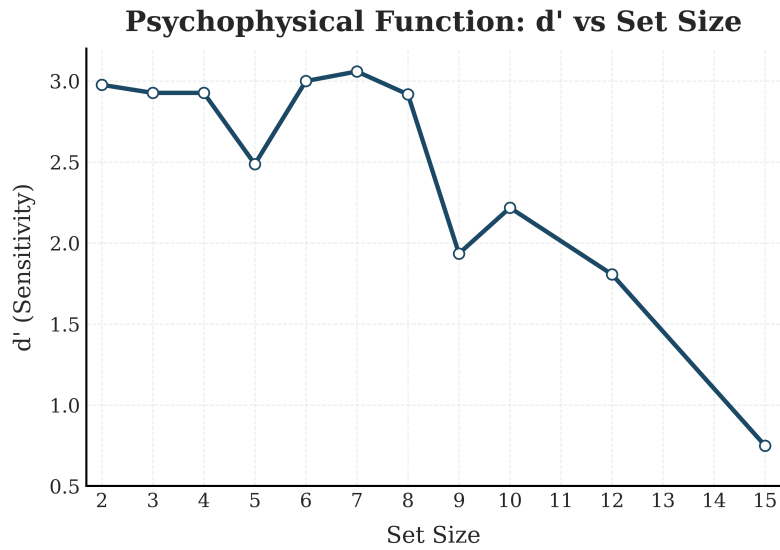


Figure 2: Psychophysical function showing the variation of sensitivity (d') with set size. As the number of words in the memory set increases, sensitivity (d') decreases, indicating reduced discriminability.

(c) Bias variation with set size:

Yes, bias varies with set size. My subject showed variable bias patterns as the task difficulty changed. The bias function (criterion c) shows a conservative bias (positive values) for smaller set sizes (2–5), with a peak at set size 5 ($c \approx 0.4$). This indicates participants were more likely to respond “No.” The criterion then drops and fluctuates around zero for moderate set sizes (6–10), indicating a relatively neutral response strategy. However, at larger set sizes (12–15), criterion c becomes increasingly negative, meaning participants tend to respond “Yes” more often, adopting a liberal bias as the task becomes harder.

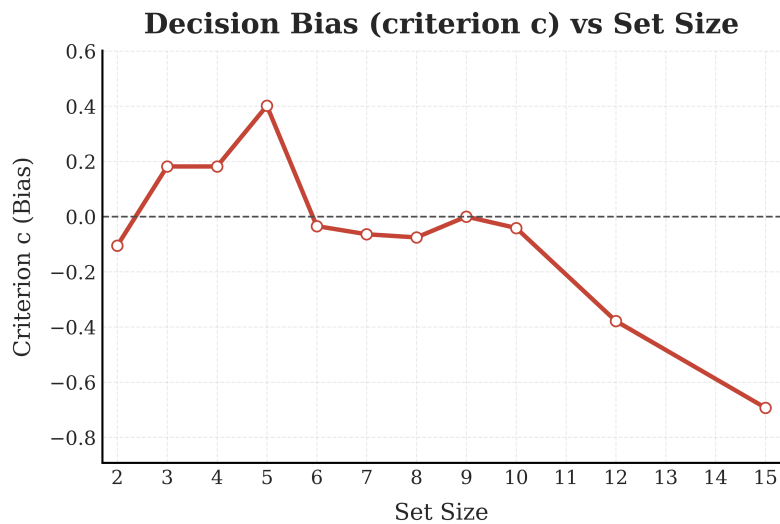


Figure 3: Variation of criterion c with set size. Positive values indicate conservative bias, negative values indicate liberal bias. The criterion shifts from conservative to liberal as set size increases, with the dashed line marking neutral bias ($c = 0$).

(d) ROC curve and relation to d' :

To construct ROC curves with varying criteria, confidence ratings were collected from the subject on each trial. These ratings allowed me to plot multiple hit rate–false alarm rate pairs by treating each confidence level as a different decision threshold.

Signal and Noise Distributions - Set Size 2 (High Sensitivity)

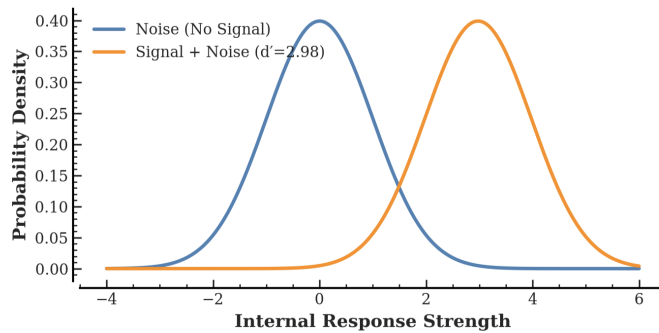


Figure 4: Signal and noise distributions for Set Size 2 (high sensitivity). The large separation between distributions ($d' \approx 2.98$) indicates strong discriminability.

For small set sizes, the noise and signal-plus-noise distributions show clear separation. My subject could easily distinguish signal-present trials, resulting in a more bowed ROC curve with high area under the curve (AUC).

Signal and Noise Distributions - Set Size 15 (Low Sensitivity)

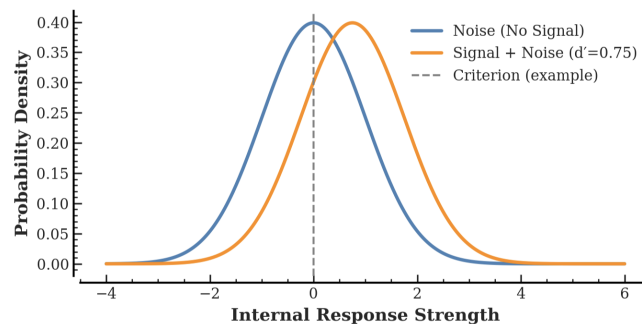


Figure 5: Signal and noise distributions for Set Size 15 (low sensitivity). The substantial overlap between distributions ($d' \approx 0.75$) indicates reduced discriminability.

At larger set sizes, the overlap between the two distributions increases significantly. My subject's internal responses for signal and noise trials became less distinct, producing a flatter ROC curve closer to the chance diagonal.

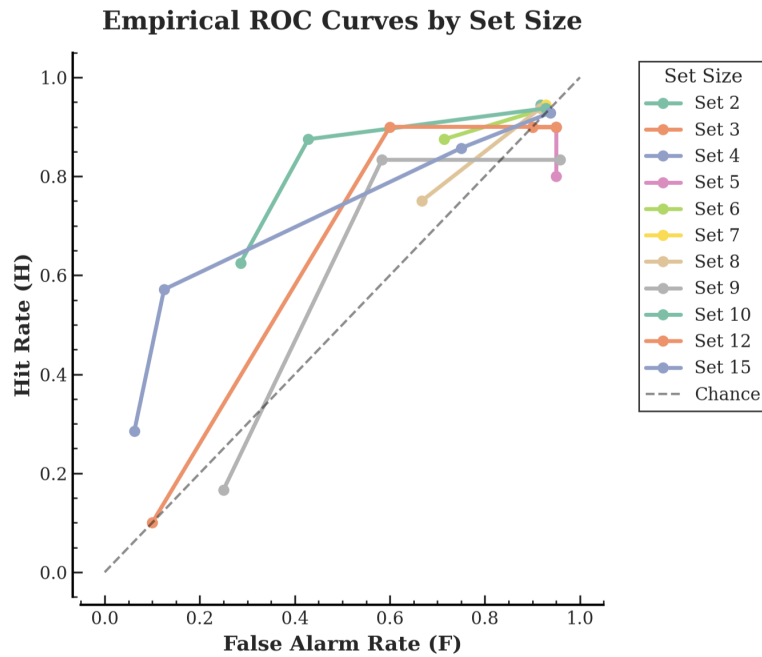


Figure 6: Empirical ROC curves for each set size. Each point represents a different confidence threshold. As set size increases, ROC curves become less bowed and approach the diagonal, indicating reduced sensitivity.

The discrete, step-wise shape of the empirical ROC curves arises because each point corresponds to one of the subject's five confidence thresholds. This pattern is typical for behavioral data collected with discrete confidence ratings (Macmillan & Creelman, 2005).

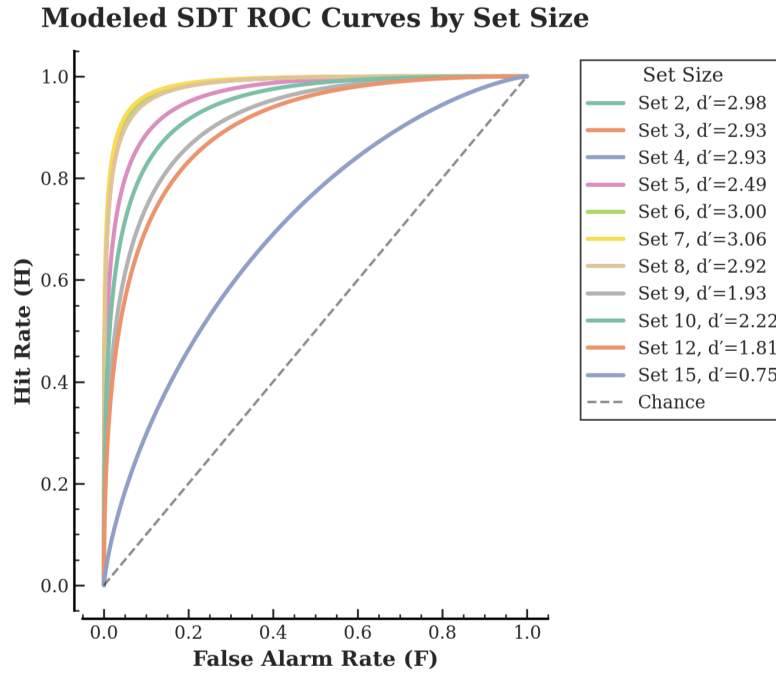


Figure 7: Modeled ROC curves generated using equal-variance SDT formulation for each set size. Larger d' values produce more bowed curves (greater discriminability), while smaller d' values produce flatter curves closer to chance.

The relationship between ROC curves and d' is clearly demonstrated: as sensitivity decreases with increasing set size, the ROC curves progressively flatten, moving from highly bowed curves (high d') toward the diagonal line representing chance performance (low d'). This direct correspondence validates the Signal Detection Theory framework for understanding working memory performance.

2 Assignment 2 (50 Points)

Questions

Test your (friend's) ESP

Can you prove or disprove parapsychology using *Signal Detection Theory (SDT)*?

- Test at least $n = 2$ (or more if you wish) individuals.
- You must provide **conclusive scientific proof** of ESP capabilities (or lack thereof) for each person tested.
- The experimental design may differ across individuals.
- Justify all experimental and analysis choices clearly.



Figure 8: Assignment 2 question.

Answers

Experimental Design: I tested 3 of my friends for ESP capabilities using three different experimental paradigms:

1. **Zener Card Test:** Classic telepathy test using 5 organ symbols (heart, lungs, brain, eye, ear) over 25 trials.
2. **Binary Choice Test:** Simplified forced-choice paradigm with 2 options (red vs. blue, heads vs. tails, left vs. right, or circle vs. cross) over 50 trials with 50% chance level. My participant chose circle vs. cross.
3. **Image Orientation Test:** Clairvoyance test requiring participants to detect whether images were displayed upright or inverted (180° rotation) over 40 trials.

Each test employed a single-device protocol where a sender viewed the target stimulus and focused on mentally transmitting it to a receiver, who then made their guess without seeing the target. All responses were recorded, and Signal Detection Theory metrics (hits, false alarms, misses, and correct rejections) were calculated.

- **Binary Choice Test (Participant 1):**

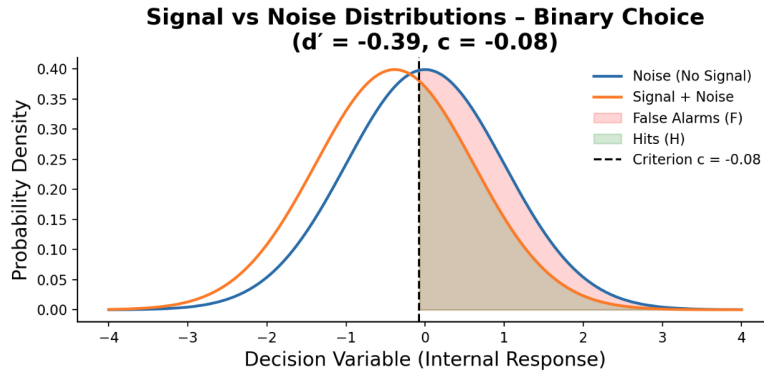


Figure 9: Signal and noise distributions in a yes–no detection task.

The Gaussian distributions represent the internal decision variable for noise (blue) and signal+noise (orange) trials. The vertical dashed line marks the observer's criterion ($c = -0.08$). Shaded regions indicate false alarms (red) and hits (green). The strong overlap between the two distributions ($d' = -0.39$) reflects low discriminability, consistent with near-chance performance.

Table 2: Signal Detection Theory metrics for Binary Choice Task.

Variable	Symbol	Value
Hit Rate	H	0.45
False Alarm Rate	F	0.61
Sensitivity	d'	-0.39
Criterion (bias)	c	-0.08
Likelihood Ratio	β	1.06
Trials (total)	N	50
Accuracy	$p(\text{correct})$	0.42
95% CI (Accuracy)		[0.29, 0.56]

In the Binary Choice (Yes/No) task, the participant achieved 42% accuracy (95% CI [0.29–0.56]), which is statistically indistinguishable from chance. The hit rate

(0.45) and false-alarm rate (0.61) resulted in a sensitivity of $d' = -0.39$, indicating almost no discriminability between signal and noise. The criterion ($c = -0.08$) and likelihood ratio ($\beta = 1.06$) both reflect a neutral decision bias. Overall, the participant's responses are consistent with random guessing, providing no evidence for ESP.

- **Zener Card Task (Participant 2, 5-AFC):**

Table 3: Confusion matrix for the 5-AFC Zener Card Test showing predicted vs. actual organ symbols.

Target	Guess				
	Brain	Ear	Eye	Heart	Lungs
Brain	2	3	2	1	1
Ear	1	0	0	0	1
Eye	3	1	2	1	0
Heart	1	1	0	1	3
Lungs	1	0	0	0	0

The confusion matrix displays my subject's response pattern across all five organ symbols. Perfect ESP performance would show concentration along the diagonal (correct predictions), while chance performance exhibits uniform distribution across all cells. The participant achieved 5 correct predictions out of 25 trials (diagonal sum: $2+0+2+1+0 = 5$), yielding exactly 20% accuracy—precisely matching the chance level for a 5-alternative forced-choice paradigm. The observed pattern shows no systematic bias toward correct identification of any particular symbol. The exact binomial test ($p = 1.000$, 95% CI [0.089–0.391]) confirms that performance is statistically indistinguishable from chance, providing no evidence for ESP capabilities.

- **Image Orientation Test (Participant 3):**

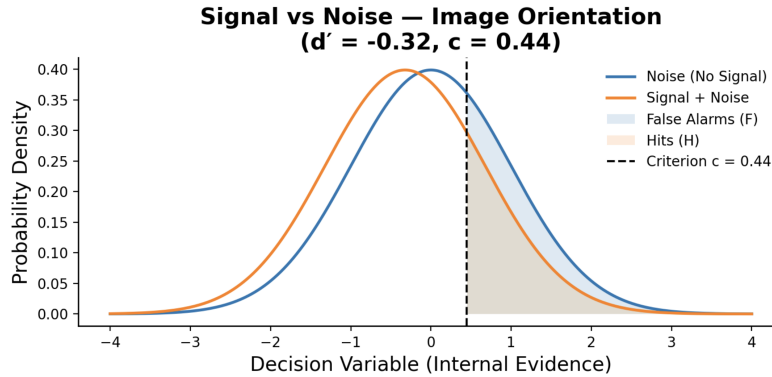


Figure 10: Signal and noise distributions in the Image Orientation (Yes/No) task.

The Gaussian distributions represent my subject's internal evidence for "upright" (signal-present) versus "inverted" (signal-absent) image trials. The overlap between the distributions is substantial ($d' = -0.32$), indicating low discriminability between the two categories. The criterion ($c = 0.44$) lies to the right of the midpoint, showing a slightly *conservative bias* — the participant tended to respond "upright" only when confident. Shaded regions indicate false alarms (blue) and hits (orange), which correspond to the decision areas in the equal-variance signal detection model.

ROC — Image Orientation (Subject 3)

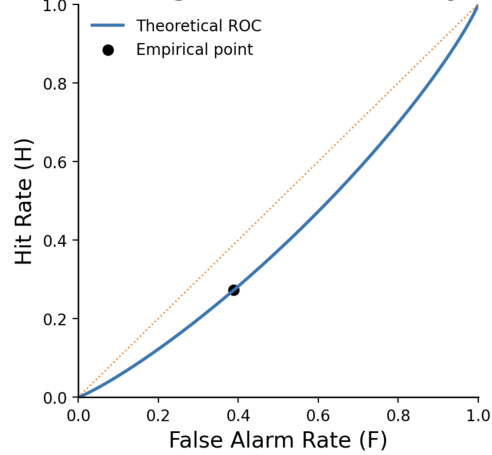


Figure 11: Receiver Operating Characteristic (ROC) curve for the Image Orientation task.

The ROC curve shows the relationship between hit rate ($H = 0.27$) and false-alarm rate ($F = 0.39$). The empirical operating point (black dot) lies close to the chance diagonal, consistent with random performance. The equal-variance SDT model (blue line) accurately captures the observed pattern, confirming that the participant's responses were statistically indistinguishable from chance. Overall, $d' = -0.32$, $c = 0.44$, and accuracy = 0.43 (95% CI [0.29–0.57], $p = 0.43$) provide no evidence for ESP or clairvoyant ability.

Acknowledgements

I would like to thank the following individuals for their contributions:

- Dr. Pavithra R, MBBS for helping me in conducting the ESP experiments and being a subject for one of the tests.
- Dr. Gnaani, MBBS, MMST for being a subject in the working memory task and providing valuable feedback on the experimental design.
- Master Vidyuth Hariharan for being a subject for one of the tests.
- Mr. Rogan Sam, B.COM, MBA for being a subject for one of the tests.

Declaration about AI Assistance

- **Claude:** Assisted with generating the HTML code for the experimental setup, Python code for computing Signal Detection Theory variables, LaTeX formatting, and writing (including spell-checking and grammar correction).
- **ChatGPT:** Assisted with generating the HTML code for the experimental setup, Python code for computing Signal Detection Theory variables, LaTeX formatting, and writing (including spell-checking and grammar correction).
- **GitHub Copilot (in VS Code):** Assisted with generating the HTML code for the experimental setup, Python code, and LaTeX formatting.

A Appendix

Stimuli Used in Experimental Tasks

The following stimuli were used across the four experimental tasks described in this study.

Working Memory Task

The following 150 words were used as the pool for the working memory task. A random subset of these words was selected for each trial.

apple, chair, water, phone, book, light, music, cloud, stone, plant, clock, bread, horse, paper, glass, metal, beach, sword, crown, flame, bridge, tower, garden, mirror, shadow, crystal, dragon, forest, ocean, desert, mountain, valley, window, pencil, bottle, camera, flower, jacket, ladder, needle, orange, pocket, rabbit, silver, table, violin, wallet, yellow, zebra, anchor, butter, candle, doctor, engine, finger, guitar, thunder, castle, diamond, eagle, feather, galaxy, hammer, island, jungle, knight, leopard, marble, neutron, oxygen, pirate, quantum, rocket, spider, temple, unicorn, volcano, wizard, angel, bronze, coffee, dancer, empire, falcon, glacier, hunter, iceberg, jester, kernel, lizard, magnet, nectar, oracle, plasma, quartz, ranger, sphere, tunnel, velvet, whisper, arrow, balloon, compass, dolphin, emerald, fountain, goblin, helmet, insect, journey, kitchen, lantern, monster, network, octopus, pyramid, rainbow, saturn, treasure, umbrella, vampire, warrior, airplane, basket, cartoon, diamond, elephant, factory, garbage, highway, icecream, jewelry, keyboard, library, machine, necklace, officer, package, quarter, railway, sandwich, teacher, uniform, village, weather, blanket, chicken, dentist, evening, freedom, goldfish, holiday, internet, justice, kangaroo, lemon, medicine, notebook, ostrich, painting, question, restaurant, skeleton, triangle, universe, vitamin, windmill, yogurt, zipper, battery, calendar, dinosaur, elevator, football, gorilla, hospital.

Binary Choice Task

Participants could select from four different binary choice types:

- **Color:** Red vs. Blue
- **Coin:** Heads vs. Tails
- **Direction:** Left vs. Right
- **Shape:** Circle vs. Cross

Zener Card Task

The Zener Card task utilized five symbolic stimuli representing human organs:

- Heart
- Lungs
- Brain
- Eye
- Ear

Image Orientation Task

The Image Orientation task employed three categories of emoji stimuli, with participants selecting one category for their session:

Animals: Lion, Elephant, Butterfly, Turtle, Octopus, Panda, Dolphin, Owl

Faces: Smiling Face, Cool Face, Thinking Face, Sleeping Face, Party Face, Nerd Face, Angel Face, Upside-Down Face

Symbols: Star, Diamond, Fire, Lightning, Moon, Sun, Rainbow, Sparkles

Mixed: Lion, Tree, Palette, Smiling Face, Star, Elephant, Flower, Guitar

Each image could be presented in either upright or inverted (180° rotated) orientation.